CookAnything: A Framework for Flexible and Consistent Multi-Step Recipe Image Generation

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Abstract

Generating instructional image sequences from recipes requires step-wise semantic disentanglement, temporal coherence, and ingredient continuity—capabilities beyond single-image diffusion models. We introduce CookAnything, a diffusion transformer framework that jointly generates an arbitrary-length sequence of step images in one denoising pass. Our approach pairs each textual step with a distinct latent region via Step-wise Regional Control, and maintains global coherence by interpolating regional and whole-recipe latents. Flexible positional encoding stabilizes long, tall canvases by resetting positional indices per step/region to avoid spatial entanglement. Cross-Step Consistency Control fuses contextual step tokens (derived from the concatenated recipe with an auxiliary cooking agent) with per-step tokens to preserve ingredient appearance across steps. On RecipeGen and VGSI-Recipe, CookAnything achieves state-of-the-art results across goal faithfulness, step faithfulness, ingredient accuracy, cross-step consistency, and usability, including 30.59 GF, 30.45 SF(C), 8.69 SF(G), and 0.19 CSC on RecipeGen (training-based), and 31.22 GF and 1.67 CSC on VGSI-Recipe (training-free). Ablations show Flexible positional encoding and CSCC are critical for consistency and alignment.

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1 Introduction

Cooking recipes describe a sequence of fine-grained actions with evolving objects and ingredients. Translating such instructions into images requires models to simultaneously achieve: (1) step-wise semantic disentanglement so that each image reflects its specific instruction; (2) temporal and spatial coherence so the scene layout remains stable across steps; and (3) ingredient continuity so entities retain identity and appearance over time. Most text-to-image diffusion models—both U-Net and DiT variants—are optimized for single images and struggle with long, multi-step prompts concatenated into one caption, often leading to semantic entanglement, repetition, or ingredient inaccuracies.

Prior attempts address related aspects: story or multi-image generation pipelines [3, 6] enforce cross-image consistency but typically assume fixed step counts or rely on sequential generation. Layout-aware conditioning [2, 4] improves spatial control but does not ensure temporal stability across steps. Recent DiT backbones such as SD3.5 [5] and FLUX [1] improve image fidelity, yet naïvely concatenating step prompts remains brittle.

We propose CookAnything, a DiT-based framework that jointly generates an arbitrary-length sequence of instruction images in a single denoising pass. The key idea is to bind each textual step to its own latent region with step-aware positional encoding, while injecting a global recipe context to maintain continuity. Our contributions are:

- A unified diffusion framework that jointly generates any number of step images in one pass with flexible canvases. - Step-wise Regional Control and flexible positional encoding for step–region disentanglement and stable spatial alignment. - Cross-Step Consistency Control that blends contextual and per-step tokens to preserve ingredient continuity. - State-of-the-art results on RecipeGen and VGSI-Recipe in both training-free and training-based settings, with strong human and automatic evaluations.

Figure 1 overviews the architecture; Figure 2 illustrates single-pass, multi-step generation.

2 Related Work

Story and multi-image diffusion seek cross-image coherence, often with sequential decoding or fixed-length schedules that accumulate errors and limit scalability [3, 6]. Layout- and grounding-aware methods improve spatial control but not temporal stability across steps [2, 4]. Beyond these, diffusion transformers with higher-capacity backbones raise fidelity but still entangle multi-step prompts when decoded as a single caption [1, 5]. Our formulation differs from story pipelines [3, 6] by decoding all steps jointly, avoiding exposure bias and fixed-length constraints. Compared to grounding methods [2, 4], we do not require external layouts; instead we bind each step's text to a latent region with step-aware positions, yielding spatial disentanglement and temporal stability without additional annotations.

3 Method

CookAnything combines three parts. Step-wise Regional Control maps each step's tokens to its latent window via regional attention plus a light base branch that anchors background and persistent objects. Flexible positional encoding resets or offsets positions per region to avoid long-range interference on tall canvases. Cross-Step Consistency Control blends recipe-level context with per-step tokens through gated cross-attention to preserve ingredient identity while keeping step specificity. Canvas. We allocate one latent window per step on a tall canvas and use regional attention so tokens attend primarily within-window while a thin global branch

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carries background cues. Positions are reset per window to limit interference across distant steps. Consistency. CSCC mixes recipe-level and per-step tokens through a learned gate conditioned on step index; it reduces ingredient drift without oversmoothing. Inference supports arbitrary step counts by resizing the canvas and running a single denoising pass.

Regional Attention. We use block-diagonal attention with a thin global band: tokens attend within their step window and to a small shared cache carrying background cues. Masks are derived from step boundaries at runtime, so the design scales to arbitrary step counts without retraining.

Flexible Positions. Positions are indexed locally inside each region and blended with a step offset; the blend weight is learned per layer via a two-way gate. This reduces long-range interference on tall canvases while preserving soft anchors for alignment.

Consistency Control. CSCC keeps two cross-attention streams—recipe context and current step—combined by a sigmoid gate conditioned on step id and token type. During training we randomly drop each stream with small probability to improve robustness.

Optimization. We fine-tune with v-parameterization and classifier-free guidance; regional windows are jittered and aspect ratios varied to regularize tiling.

Architecture and Training Details. We use a DiT backbone with shared weights across a full-width main stream and a 1/4-width global stream; regional routing is realized with token adapters and attention masks so parameters remain near the base model. Classifier-free guidance is scheduled along the trajectory: linearly warm up $1.0 \rightarrow 3.0$ over the first 30% denoising steps, then cosine-decay to 2.0 to avoid early over-conditioning. Sampling follows v-parameterized EDM with Karras ρ=7; in training-free mode we retain the backbone sampler and only apply regional masks and CSCC gates. A lightweight cooking agent extracts entities (ingredients, tools) and normalizes aliases; its summary is concatenated with each step's text before tokenization. We cap 64 tokens per step and 512 globally to bound latency on long recipes. Data augmentation includes step-count jitter, neutral-step permutations, aspect jitter, and light mask noise; regularizers include region drop and context drop to improve robustness to omissions and insertions.

4 Experiments

We evaluate on RecipeGen and VGSI-Recipe using goal/step faithfulness, ingredient accuracy, cross-step consistency, and usability. Baselines include story diffusion, grounded diffusion with layout control, and strong DiT backbones, in both training-free and fine-tuned modes. Training-free attaches our modules to frozen backbones; training-based fine-tunes on the same splits. CookAnything consistently improves faithfulness and lowers consistency errors; examples appear in Figures 5 and 2. We report goal/step faithfulness via text-image similarity, ingredient accuracy via entity matching, and cross-step consistency via track-aware feature distance; human studies rate alignment and continuity on 5-point scales.

5 Analysis

Ablations show flexible positions are key for spatial stability, while CSCC curbs identity drift and omissions. Joint single-pass decoding avoids autoregressive error buildup and reduces latency versus per-step decoding. Long or ambiguous recipes still induce minor layout jitter, suggesting stronger global context is beneficial.

6 Conclusion

CookAnything introduces a unified diffusion framework for arbitrary-length, joint generation of recipe step images. By coupling step-aware regional control, flexible positional encoding, and cross-step consistency control, the method delivers flexible, coherent, and ingredient-faithful sequences. Experiments on RecipeGen and VGSI-Recipe demonstrate state-of-the-art results across automatic and human evaluations. By enabling arbitrary-length, single-pass generation with stable layouts and ingredient continuity, CookAnything provides a practical building block for instruction-following visual content beyond cooking recipes.

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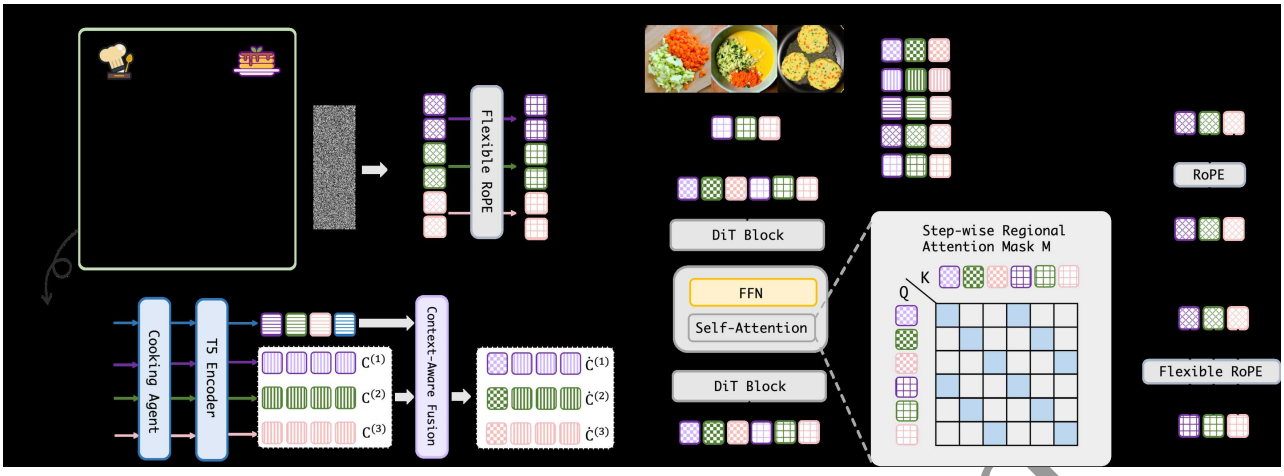

Figure 1: Overview of CookAnything: a DiT-based diffusion framework that jointly generates an arbitrary-length sequence of step images. Step-wise Regional Control assigns step texts to latent regions with regional attention masks; flexible positional encoding stabilizes long, tall canvases; CSCC fuses contextual and per-step tokens for ingredient continuity.

Figure 2: Single-pass multi-step generation on a sample recipe. The model produces all step images jointly, preserving spatial layout and ingredient continuity across steps.

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Figure 5: Qualitative comparisons with baselines. CookAnything maintains coherent layouts and consistent ingredient appearance across steps.

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